

SUPPLEMENTARY PROBLEM: ASYMPTOTIC NOTATIONS

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Problem 1

Considering functions $f(n) \geq 0$, $g(n) \geq 0$, and constant $c > 0$, indicate whether each of the following statements is true. Prove the statements that are true by providing a formal argument that is based on the definition of asymptotic notation. For statements that are false, provide a counter-example to prove that they are false.

- (a) $\max\{f(n), g(n)\} = \Theta(f(n) + g(n))$.
- (b) $f(n) + c = O(f(n))$.
- (c) $f(2n) = \Omega(f(n))$.
- (d) If $f(n) = O(g(n))$, $\log(f(n)) \geq 0$, and $\log(g(n)) \geq 1$, then $\log(f(n)) = O(\log(g(n)))$

Problem 1 Solutions

- (a) $\max\{f(n), g(n)\} = \Theta(f(n) + g(n))$.

Proof. We need to prove two claims:

- **Case 1:** $\max\{f(n), g(n)\} = O(f(n) + g(n))$:

We need to show that $\max\{f(n), g(n)\} \leq d(f(n) + g(n))$ for some constant $d > 0$.

We can decompose $f(n) + g(n)$ such that:

$$\begin{aligned} \max\{f(n), g(n)\} &\leq d(f(n) + g(n)) = d(\max\{f(n), g(n)\} + \min\{f(n), g(n)\}) \\ 0 &\leq (d - 1) \max\{f(n), g(n)\} + d \min\{f(n), g(n)\} \end{aligned} \quad (1)$$

Since both $f(n)$ and $g(n)$ are non-negative Case 1 is satisfied for $d = 1$.

- **Case 2:** $\max\{f(n), g(n)\} = \Omega(f(n) + g(n))$

We need to show that $\max\{f(n), g(n)\} \geq d(f(n) + g(n))$ for some constant $d > 0$.

For any two real numbers, the average is lesser than or equal to the maximum. Thus,

$$\max\{f(n), g(n)\} \geq \frac{f(n) + g(n)}{2} \quad (2)$$

Hence, this case is satisfied for $d = 0.5$.

□

$$(b) f(n) + c = O(f(n)).$$

Disproof. If this statement is true, then there would have to be a constant $d > 0$ such that $f(n) + c \leq d * f(n)$ for all $n > n_0$ for a constant $n_0 > 0$. Let $f(n) = 0$. Then, it follows that $c \leq 0$. However, this is a contradiction since $c > 0$. □

$$(c) f(2n) = \Omega(f(n))$$

Proof. We can make the similar observations and prove that this claim is false by a counter example. In this case, we should choose again an exponential function where exponent becomes smaller as the input increases. When $f(n) = e^{-n}$, then $f(2n) = e^{-2n}$. We can write these into the Ω definition.

$$\begin{aligned} f(2n) &\geq cf(n) \\ e^{-2n} &\geq ce^{-n} \\ e^{-n} &\geq c \end{aligned}$$

Since $\lim_{n \rightarrow \infty} e^{-n} = 0$, given some $c > 0$, there will always exist some $n \geq n_0$ s.t. the inequality does not hold true. Therefore, the given statement is false.

□

$$(d) \text{ If } f(n) = O(g(n)), \log(f(n)) \geq 0, \text{ and } \log(g(n)) \geq 1, \text{ then } \log(f(n)) = O(\log(g(n)))$$

Proof. If $f(n) = O(g(n))$, then there is a constant $d > 0$ such that $f(n) \leq g(n)d$. Thus,

$$\begin{aligned} f(n) &\leq g(n)d \\ \log(f(n)) &\leq \log(g(n)) + \log(d) \end{aligned} \tag{3}$$

We need to show that $\log(f(n)) \leq \log(g(n))k$ for some $k > 0$. Let's consider two cases:

$$- \log(d) \leq 0:$$

$$\log(f(n)) \leq \log(g(n)) + \log(d) \leq \log(g(n)) \tag{4}$$

Thus, this case is satisfied for $k = 1$.

$$- \log(d) > 0: \text{ As } \log(g(n)) \geq 1,$$

$$\begin{aligned} \log(f(n)) &\leq \log(g(n)) + \log(d) \\ \log(f(n)) &\leq \log(g(n)) + \log(d) \log(g(n)) \\ \log(f(n)) &\leq \log(g(n))(1 + \log(d)) \end{aligned} \tag{5}$$

Thus, this case is satisfied for $k = 1 + \log(d)$ which is greater than 0.

□